João Gabriel de Barros Rocha

Software Engineer

Vitória, Brazil (Open to Relocation) | +55 27 99972-1208 | jgabriel.b.rocha@gmail.com
linkedin.com/in/joaogbarros | github.com/JoaoGBarros

Professional Summary

Software Engineer who builds web applications end to end: REST APIs and backend services, SQL databases and React/TypeScript frontends, graduating with a BSc in Computer Engineering in December 2026. As the sole backend and infrastructure engineer at a Spanish proptech startup, I own the Python 3 (FastAPI) web service, the PostgreSQL database and the external integrations (Stripe, Mapbox, LLM APIs) behind a web platform of 100,000+ listings. Experienced with Docker, Git branching workflows with pull-request reviews and CI/CD, and a daily user of AI coding agents within a disciplined, review-driven workflow. Fluent in English (C2) and used to working in multidisciplinary, agile teams.

Skills

Languages: TypeScript, JavaScript, Python 3, Java, SQL, Kotlin, PHP, C++
Frontend: React, Vue.js, HTML, CSS, Angular, Electron (Node.js), UI implementation with designers, web design principles
Backend & APIs: REST APIs, FastAPI, Flask, Spring Boot, Jakarta EE, WebSockets, JPA/ORM, Microservices, Event-Driven Architecture, Apache Kafka, third-party API integration (Stripe, Mapbox, LLM APIs)
Databases: PostgreSQL (schema design, provisioning, indexing, query tuning, PostGIS/GIST), MySQL, Redis
DevOps & Cloud: Docker, Docker Compose, Linux, Nginx, Git, GitHub Actions, GitLab CI/CD, Hetzner, Azure, AWS, Google Cloud, Cloudflare
AI: AI coding agents (Claude Code, Cursor), LLM integration (Claude, GPT), Computer Vision for image filtering
Practices: Unit, Integration & E2E Testing (Pytest, Protractor), Code Review, Application Security, On-call & Incident Response, Scrum

Professional Experience

Property Guys, Spanish Proptech Startup Mar 2026 -- Present
Back-end Software Engineer (Contract) *Remote, Spain*

- Sole software engineer on the backend and infrastructure of a production real-estate web platform for the Mallorca market, with 100,000+ listings aggregated from roughly 1,000 agency websites; I own the FastAPI web service and REST API, the crawler, servers, networking and deployment in a remote, English-speaking Scrum team.
- Restructured the inherited monolith into Router-Service-Repository layers and fixed SQL injection, account-takeover and user-enumeration vulnerabilities.
- Integrated the platform with external systems: built the Stripe subscription flow end to end (checkout, plans, billing portal, webhooks), Mapbox geocoding for location validation and LLM APIs for listing descriptions, plus lead routing and image-removal controls for subscribed agencies.
- Cut homepage load time by 84% and brought search to about 250 ms per page through PostgreSQL work: B-tree and GIST spatial indexes, materialized views, N+1 query fixes and moving ranking logic into SQL.
- Migrated production from Azure to Hetzner, provisioning the servers, the PostgreSQL instance, DNS and security, and cut hosting costs by roughly 75%.
- Set up a Git workflow with protected branches and pull-request reviews, CI/CD in GitHub Actions with automatic rollback when integration tests fail, Pytest unit tests and daily database backups.
- Unified the company's two crawlers, my Scrapy pipeline and an LLM-based crawler that ran about 65 unversioned cron jobs on the server, into a single versioned repository with a shared database write contract and automated deploys.
- Designed the data-quality guards: sold and inactive listing detection, location validation with a geocoding fallback, price recovery from JavaScript-rendered sites with Playwright, and cross-agency duplicate detection through image fingerprinting (pHash).
- Reduced hallucinations in LLM-generated listing descriptions and added a computer-vision step that removes stock photos, logos and agent portraits before publication; maintained the platform's HTML, CSS and JavaScript email templates.

Software Engineering Practices Laboratory (LabES), UFES Aug 2023 -- Mar 2026
Junior Software Engineer *Vitória, Brazil*

- Software engineer on ES+ Criativo and HUB ES+, production web applications for the Espírito Santo State Secretariat of Culture that have supported 1,400+ events.
- Built the first version of both systems in Java (Jakarta EE, JPA, JSF), from the relational data model and backend services to the web UI, then led the platform's migration to PHP and WordPress, writing custom plugins for features outside WordPress core.

- Worked in a multidisciplinary team with designers and stakeholders, leading delivery meetings with the Secretariat and turning client needs into technical specifications and interface requirements.
- Served as the team's on-call engineer for production incidents and covered the full lifecycle: requirements, technical documentation, code review, deployment and user guides for non-technical staff.

Vixteam Consultoria & Sistemas Dec 2021 -- Feb 2023
Software Development Intern *Vitória, Brazil*

- Built the end-to-end test suite for Team Audit, Vixteam's audit-management web platform, with Protractor and TypeScript, raising test coverage and catching regressions before each release.
- Fixed bugs across the full stack of this large-scale web application, from Angular and TypeScript components to Spring Boot services and REST APIs.

Projects

CaLMo, Causal Loop Diagram Modeler (Best Paper, ERI-ES 2026) 2026
Research Project, NEMO, UFES: Vue.js, Python (Flask), PostgreSQL, Kafka, WebSockets *dev.nemo.inf.ufes.br/calmo*

- Evolved CaLMo, a web tool for Causal Loop Diagram modeling, with an interactive graphical editor in Vue.js and vis.js (MVVM), delay-aware archetype detection, thematic subsystems and import/export in JSON, CSV and XMILE.
- Added real-time collaborative editing: WebSocket synchronization via Flask-SocketIO and an Apache Kafka broker that orders every change and records each diagram's edit history, extending the Flask REST API and PostgreSQL schema; deployed the new version on the research group's server.

Here To Slay, Distributed Multiplayer Card Game 2026
Personal Project: React, TypeScript, Java 21, Apache Kafka, Docker *github.com/JoaoGBarros/HereToSlay*

- Developed the client in React and TypeScript on Electron (Node.js), connected to the backend through a WebSocket gateway, and containerized the full stack with Docker Compose behind an Nginx reverse proxy.
- Built an event-driven backend where gateway, lobby and game-engine services communicate only through an Apache Kafka cluster (3 brokers, KRaft mode), with event sourcing and crash recovery through compacted topics.

CT9 ''Na Palma da Mão'', Academic Management Platform 2024
Personal Project: Java, Spring Boot, PostgreSQL, Docker, Angular, Kotlin *github.com/JoaoGBarros/CT9-Backend*

- Designed the PostgreSQL schema from scratch and built the RESTful Spring Boot backend with role-based authentication, containerized with Docker on Google Cloud Run and serving an Angular web client and a Kotlin Android app.

Education

Federal University of Espírito Santo (UFES) Expected December 2026
BSc in Computer Engineering Vitória, Brazil

Publications & Awards

Best Paper Award, Escola Regional de Informática do Espírito Santo (ERI-ES 2026), for: J. G. de Barros Rocha, J. de Souza Borges, T. F. N. Lahass, A. B. Apolinário, M. P. Barcellos. *Evolving CaLMo to Enhance Causal Loop Diagram Modeling: New Elements, Interactive Graphical Support, Collaboration, and Interoperability*. Proceedings of ERI-ES 2026 (to appear).

Languages

Portuguese (Native) | English (C2): advanced immersion at ILAC Vancouver, Canada; Cambridge C1 Advanced (CAE)

Leadership & Volunteering

IT Academic Center, UFES Aug 2023 -- Nov 2024
Student Ombudsman Coordinator *Vitória, Brazil*

- Elected to represent 500+ Computer Science and Engineering students, negotiating directly with department heads to secure a dedicated student lounge and funding for a building renovation.
- Co-organized the 9th Regional School of Informatics with the Brazilian Computer Society (SBC), managing logistics and promotion for an event with 120+ attendees.

Introcomp Apr 2020 -- Feb 2022
Volunteer, Content, Tutoring and Data Science *Vitória, Brazil*

- Taught Python programming fundamentals to public high school students and analyzed student performance data with Pandas and Matplotlib.

Network Administration Lab, UFES Mar 2019 -- Jun 2019
Volunteer, Technical Support and Scripts *Vitória, Brazil*

- Created Bash automation scripts for Linux systems and supported students with access and machine issues.